

Applications: Growth and Decay

ANSWER KEY



Growth models

- 1 A city's population is modeled by $P(t) = 12000(1.035)^t$, where t is in years.
- a State the initial population and the annual growth rate. Initial population 12,000; growth rate 3.5% per year.
- b Find the population after 10 years, to the nearest person. $P(10) = 12000(1.035)^{10} \approx 16,927$.
- 2 An account earns 4% annual interest compounded continuously. How long does it take an investment to double? Give an exact expression and a decimal to two places.
- $2 = e^{0.04t}$, so $t = \frac{\ln 2}{0.04} \approx 17.33$ years.
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Decay models

- 3 Carbon-14 has a half-life of 5730 years: $A(t) = A_0\left(\frac{1}{2}\right)^{t/5730}$.
- a What fraction of the original carbon-14 remains after 2000 years? Answer to three decimal places.
- $\left(\frac{1}{2}\right)^{2000/5730} \approx 0.785$.
- b A sample retains 30% of its carbon-14. Estimate its age to the nearest hundred years.
- $0.3 = (0.5)^{t/5730}$ gives $t = 5730 \frac{\ln 0.3}{\ln 0.5} \approx 10,000$ years.
- 4 A 200 g sample of a radioactive substance decays to 50 g in 12 hours, following $A(t) = 200e^{kt}$.
- a Find the decay constant k (4 d.p.). $\frac{50}{200} = e^{12k}$, so $k = \frac{\ln(1/4)}{12} \approx -0.1155$.
- b Find the half-life of the substance. $200 \rightarrow 50$ is two halvings in 12 h, so the half-life is exactly 6 hours (equivalently $\frac{\ln 2}{0.1155} \approx 6.0$ h).
- 5 Newton's law of cooling: coffee at 90°C cools in a 20°C room following $T(t) = 20 + 70e^{-0.1t}$ (t in minutes). Find the temperature after 10 minutes, to one decimal place.
- $T(10) = 20 + 70e^{-1} \approx 20 + 25.75 = 45.8^\circ\text{C}$.
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Logistic growth (an introduction)

- 6 A lake is stocked with fish, and the population is modeled by $P(t) = \frac{5000}{1 + 9e^{-0.4t}}$, where t is in years.
- a Find the initial population $P(0)$. $P(0) = \frac{5000}{1+9} = 500$.
- b Find the carrying capacity (the value P approaches as $t \rightarrow \infty$). As $t \rightarrow \infty$, $e^{-0.4t} \rightarrow 0$, so $P \rightarrow \frac{5000}{1} = 5000$.
- c Find $P(10)$, to the nearest fish. $P(10) = \frac{5000}{1 + 9e^{-4}} \approx \frac{5000}{1.1648} \approx 4292$.

Comparing models

- 7 Two towns start with 10,000 people. Town A grows at a constant rate of 300 people/year (linear); Town B grows at 3% per year (exponential). Find each town's population after 15 years, to the nearest person, and state which is larger.

Town A: $10000 + 300(15) = 14,500$. Town B: $10000(1.03)^{15} \approx 10000(1.5580) \approx 15,580$. Town B is larger.

- 8 A radioactive isotope decays from 80 g to 10 g over 3 half-lives. Find the half-life in terms of the decay time, and write the exponential decay model $A(t) = 80e^{kt}$, finding k if the total elapsed time is 15 years (4 d.p.).

$80 \rightarrow 10$ over 3 halvings ($80 \rightarrow 40 \rightarrow 20 \rightarrow 10$) means 3 half-lives = 15 years, so half-life = 5 years.

Using $A(15) = 10$: $10 = 80e^{15k}$, so $k = \frac{\ln(1/8)}{15} \approx -0.1386$.